

DECIDE Manual

Version 1.2

1 October 2026

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Contents

Part A -- The DECIDE Expression Language 4
- Expression engine . . . 4
- Operators . . . 4
- Scalar functions . . . 4
- Probability and summary functions . . . 6
- Distribution functions . . . 6
- Distribution types and parameters . . . 7
- Tables and Lookup . . . 9
- Table and Lookup examples . . . 9
- TableToProb and TableToTime . . . 10
- Parameters vs. Variables . . . 10

Part B -- Decision Tree 11
- File panel . . . 11
- Decision Tree Editor . . . 11
- Model Calibration . . . 13
- Algorithms . . . 13
- Bayesian Calibration . . . 13
- Applying a result . . . 13
- PSA Setup . . . 14
- Correlation . . . 14
- Correlated Distribution . . . 14
- Value of Information (EVPI/EVPPI) . . . 15
- Population EVPI/EVPPI . . . 15

Part C -- Markov Model 16
- Engine variables . . . 16
- Markov tenure states . . . 16
- Markov Editor . . . 16
- File panel . . . 16
- Model Calibration . . . 16
- PSA Setup . . . 16
- Strategy comparison and ICER . . . 17
- Value of Information (EVPI/EVPPI) . . . 17

Part D -- Partitioned Survival Model (PSM) 18
- How a PSM node is calculated . . . 18
- File panel . . . 18
- PSM Editor . . . 18
- Survival distributions . . . 19
- Fit from KM / patient data . . . 19

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Curve tools: library, extrapolation and plausibility . . . 19
Model properties . . . 20
Validation . . . 20
Base case results . . . 20
Survival Curves / State / Time Report . . . 20
Result Dashboard . . . 21
Documentation exports . . . 21
Validation self-checks . . . 21
Export code . . . 22
Model Calibration, Sensitivity Analysis, PSA and Value of Information . . . 22
Worked example . . . 22
Part E -- Microsimulation 23
How a Microsim is calculated . . . 23
Health states and branch nodes . . . 23
File panel . . . 24
Microsim Editor . . . 24
Model properties . . . 24
Trackers . . . 25
Resampling distributions . . . 25
Validation . . . 25
Run and Base case results . . . 26
Patient Journey . . . 26
Documentation exports . . . 26
Validation self-checks . . . 27
Sensitivity Analysis, PSA, Dashboard and Model Calibration . . . 27
Worked example . . . 27

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Part A -- The DECIDE Expression Language

The knowledge in this part is shared by every model type in DECIDE (Decision Tree, Markov, and future model types): the same expression evaluator, the same functions, and the same Parameters/Variables/Distributions/Tables data model underlie every Cost, Effect, Probability, and definition field in the app.

Expression engine

DECIDE expressions use Python-compatible semantics. The same expression rules are available in decision trees and Markov models.

Parameters are numeric inputs. Variables are named expressions that may reference parameters and other variables.

Example: `prob_AE = RateToProb(rate_AE_3yr, 1, 1/4)` with `rate_AE_3yr = 0.5` gives approximately `0.1175030974`.

Operators

Operator	Meaning
+ - * /	Addition, subtraction, multiplication, division
**	Exponentiation
% //	Modulus and floor division
== != < <= > >=	Comparisons
and or not	Logical operators
(...)	Grouping and precedence

Constants: e is Euler's number and `pi` is pi. Trigonometric functions use radians.

Scalar functions

Function	Meaning
abs(x)	Absolute value
arccos(x), arcsin(x), arctan(x)	Inverse trigonometric functions
bound(x, (a,b)) or bound(x, [a,b])	Clamps x into the closed interval [a, b]. The bracket style does not change the result; mixed forms such as (a,b] are not valid syntax.
cbrt(x)	Cube root
ceil(x), floor(x)	Ceiling and floor

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Function	Meaning
comb(n, k)	Binomial coefficient; n >= k >= 0
cos(x), sin(x), tan(x)	Trigonometric functions
cosh(x), sinh(x), tanh(x)	Hyperbolic functions
erf(x), invErf(x)	Error function and inverse error function
exp(x)	e**x
factorial(n)	Factorial for a non-negative integer
gamma(x), lgamma(x)	Gamma and log-Gamma functions
hypot(x,y)	sqrt(x**2 + y**2)
if(condition, a, b)	Return a if condition is true, otherwise b
ln(x)	Natural logarithm
log(x,b), log10(x)	Logarithm to base b and base 10
logit(p), logis(x)	Log odds and logistic function
max(a,b), min(a,b)	Maximum and minimum
round(x), round(x,n)	Python-compatible rounding
sign(x)	-1, 0, or 1 according to the sign of x
sqrt(x)	Square root

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Probability and summary functions

Function	Meaning
ProbToProb(p,t1,t2)	1 - (1-p)**(t2/t1)
ProbToRate(p,t1,t2)	-ln(1-p) * t2/t1
RateToProb(r,t1,t2)	1 - exp(-r * t2/t1)
mean(x1,...,xn)	Arithmetic mean
prod(x1,...,xn)	Product
quantile({x1,...,xn},q)	Linear-interpolated quantile, 0 <= q <= 1
sd(x1,...,xn)	Sample standard deviation, denominator n-1
sum(x1,...,xn)	Sum
var(x1,...,xn)	Sample variance, denominator n-1

Distribution functions

The distribution argument is the name of a distribution configured in the Distribution section. Its type and parameters are not repeated in the function call.

Function	Meaning
DistMean(distribution)	Mean of the configured distribution
DistSample(distribution)	Random sample from the configured distribution
DistCDF(distribution,s)	Cumulative probability P(X <= s)
DistPPF(distribution,p)	Value s whose cumulative probability is p
DistSampleCond(distribution, interval)	Conditional random sample in an interval such as [lower, upper] or (lower, upper)
DistToProb(distribution, t1,t2)	Conditional transition probability during [t1,t1+t2] given no event before t1
DistToTime(distribution, t1)	Random remaining time until the event for someone still event-free at t1 (t1 optional, default 0); measured from t1, like t2 in DistToProb
DistTrim(distribution,[min,max])	Random sample trimmed to min <= X <= max
DistSurv(distribution,s)	Survival probability, 1 - DistCDF(distribution,s)
DistHazard(distribution, s)	Hazard, PDF(s) / survival(s)

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

DistToProb is calculated from the conditional survival S(t1+t2)/S(t1), not a plain CDF or PDF. DistToTime solves the same formula for time instead of probability, so the two always agree: the share of DistToTime(dist, t1) draws that are <= t2 equals DistToProb(dist, t1, t2).

Distribution types and parameters

DECIDE follows the parameterization of the R package flexsurv (for most families also the one of R's own d/p/q/r functions, such as pweibull and qgamma). Values fitted with flexsurv, or reported in a paper that uses flexsurv, can be typed straight into the Distribution table. The Gamma distribution uses **shape and rate**, as flexsurv does (scale = 1/rate).

Distribution (parameters)	Survival function S(t) and notes (t in the distribution's time unit)
Exponential (rate)	S(t) = exp(-rate*t); mean 1/rate
Weibull (shape, scale)	S(t) = exp(-(t/scale)^shape); mean scale * Gamma(1 + 1/shape)
Gamma (shape, rate)	mean shape/rate, variance shape/rate^2; scale = 1/rate
Log-normal (meanlog, sdlog)	S(t) = 1 - Phi((ln t - meanlog)/sdlog)
Log-logistic (shape, scale)	S(t) = 1/(1 + (t/scale)^shape); median = scale
Gompertz (shape, rate)	hazard h(t) = rate*exp(shape*t); S(t) = exp(-(rate/shape)*(exp(shape*t) - 1))
Generalized gamma (mu, sigma, Q)	Prentice (flexsurv gengamma) parameterization; Q = 0 is the log-normal

Time unit. DECIDE never converts time units. The distribution's parameters must be in the same unit as the arguments of DistToProb, DistToTime, DistSurv and so on, and as the model cycle (for example, a Weibull fitted in years used with monthly cycles needs DistToProb(d, t1/12, 1/12)).

Entering values reported in other conventions. The same name can mean different parameters in different software, so convert before typing:

Source / reported as	Enter in DECIDE
R pweibull, flexsurvreg(dist="weibull"): shape, scale	As reported
flexsurvreg(dist="weibullPH"), Stata streg weibull (proportional hazards): cumulative hazard lambda * t^gamma	shape = gamma, scale = lambda^(-1/gamma)
Paper or software with cumulative hazard (lambda * t)^gamma (for example the DARTH time-dependent Sick-Sicker tutorial)	shape = gamma, scale = 1/lambda
survival::survreg Weibull (accelerated failure time): Intercept, Log(scale)	scale = exp(Intercept), shape = 1 / survreg scale

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Source / reported as	Enter in DECIDE
survreg log-logistic: Intercept, scale	scale = exp(Intercept), shape = 1 / survreg scale
survreg log-normal: Intercept, scale	meanlog = Intercept, sdlog = scale
Exponential given as a mean time theta	rate = 1/theta
Gamma given as shape k and scale theta	shape = k, rate = 1/theta
Gamma given as mean m and standard deviation s	shape = (m/s)^2, rate = m/s^2
flexsurvreg(dist="gengamma.orig"): shape b, scale a, k	Q = 1/sqrt(k), sigma = 1/(b*sqrt(k)), mu = ln(a) + ln(k)/b

With flexsurv output, use the natural-scale estimates (est in res), not the log-scale ones (res.t), and with covariates compute the parameter at the covariate values of the group being modelled before typing it. The Weibull is the most common source of confusion: a shape of 1.1 with lambda = 0.08 gives scale = 12.5 when the cumulative hazard is (lambda*t)^gamma but scale = 9.94 when it is lambda*t^gamma, and the one-cycle event probability at time 0 is 0.060 versus 0.077.

Gamma and older models. Earlier versions of DECIDE stored the Gamma distribution as k (shape) and theta (scale). Such models are converted when the Distribution row is loaded: k becomes shape and theta becomes rate = 1/theta (an expression is wrapped as 1/(expression)), and the distribution, and every result that depends on it, is unchanged. A PSA row that targets a Gamma slot by its old name theta must be pointed at rate again.

Other distributions (parameters as shown in the Distribution table):

Distribution (parameters)	Definition
Beta (a, b)	R shape1, shape2; mean a/(a+b)
Normal (mu, sigma)	Standard deviation, not variance
Truncated Normal (mu, sigma, a, b)	mu and sigma describe the untruncated normal; a, b are the bounds
Uniform (a, b)	Minimum and maximum
Triangular (a, b, c)	Minimum, mode, maximum
PERT (a, b, c)	Minimum, most likely, maximum; Beta with weight 4
Poisson (lambda)	Mean count
Binomial (n, p); Bernoulli (p)	Number of trials and probability; probability of success
Geometric (p)	Number of trials until the first success (starts at 1)
Negative Binomial (r, p)	Number of failures before the r-th success
Discrete Uniform (a, b)	Integers from a to b, both included
Hypergeometric (w, b, n)	White, black, number drawn

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Distribution (parameters)	Definition
Categorical (p)	List of probabilities that sums to 1
Student's t (nu); Chi-Square (k)	Standard t (location 0, scale 1); degrees of freedom
Cauchy (x0, gamma); Half-Cauchy (gamma)	Location and scale; scale
Half-Normal (sigma); Laplace (mu, b); Logistic (mu, s)	Scale; location and scale; location and scale
Gumbel (mu, beta)	Maximum-extreme (right-skewed) form
Pareto (alpha, k)	Shape, minimum (type I)
Zipf (s, n)	Exponent, number of elements

A life-table column is a probability or a rate depending on its label: qx is an annual probability and is used directly, mx is a rate and goes through RateToProb.

Tables and Lookup

A Table is a named grid of reference data, entered directly or uploaded from a CSV or Excel file. Lookup syntax:

Form	Reads
table_name[lookup_value]	Default result column
table_name[lookup_value, result_column]	Named/indexed result column
table_name[lookup_value, lookup_column, result_column]	Named/indexed lookup and result columns
table_name[..., method]	Optional trailing method: "exact" | "truncate" | "linear" (default) | "spline"

Table and Lookup examples

Example	Meaning
t_mortality[Current_age, "Age", "Mortality"]	Age-banded mortality, linear interpolation
t_mortality[_cycle, "Cycle", "Prob"]	A Markov transition probability that changes every cycle
t_cost[Current_age, 0, 1, "exact"]	Exact age-band cost lookup by column index

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Example	Meaning
RateToProb(t_hazard[_cycle, "Cycle", "Rate"], 1, 1)	Lookup combined with a rate/probability conversion

TableToProb and TableToTime

The table counterparts of DistToProb/DistToTime, for survival or mortality tables (e.g. a life table) instead of parametric distributions. TableToProb(table, t1, t2[, value_column[, type]]) returns the probability of the event in (t1, t1+t2] for someone event-free at t1; TableToTime(table, t1[, value_column[, type]]) returns the random remaining time from t1. The type argument accepts "conditional" (default -- q(x)), "probability", or "cumulative", depending on the input table's form.

Parameters vs. Variables

Parameter: a root input, with name / base / low / high / source. Base can be a number or a formula referencing other Parameters. Low/High feed Sensitivity Analysis and Model Calibration bounds. A Parameter can have a PSA distribution attached.

Variable: a quantity computed from other Parameters/Variables via a definition (formula). In a Markov model, a Variable's definition can depend on `_cycle`/`_tenure` (see Part C), so its value can change by cycle; a Decision Tree Variable always evaluates with `_cycle=0`, `_tenure=1`, so its value is fixed.

Parameters and Variables share one namespace -- a Variable's name cannot duplicate a Parameter's name. Use `prob_AE` in a decision-tree branch probability, terminal cost/effect, Markov initial probability, state cost/effect, or transition probability.

Renaming is automatic. Rename a Parameter, Variable, Distribution, Table or Tracker in its input table and every place that uses the old name is rewritten to the new one -- canvas formulas (branches, states, transitions), other Variables' definitions, Distribution parameters, trackers and rates, PSA setup and Sensitivity/Calibration selections -- in Decision Tree, Markov, PSM, DES, Microsimulation and ABM alike. Text inside quotes (a table column such as `tbl["age"]`) and longer names that merely contain the old one are never touched. If the new name is already taken, nothing is rewritten and the duplicate-name warning shows.

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Part B -- Decision Tree

File panel

Decision Tree, Markov, PSM, ABM and DynaTrans share the same File panel; NMA has Save / Save as / Open with its own autosave.

Control	What it does
Save	Writes the model straight to Folder/Filename and overwrites an existing file without asking. Once a file is saved or opened, Save keeps updating that file.
Save as	Downloads a copy through the browser, where you choose the name and folder.
Folder / Saved models / Open saved model	Choose the folder Save and Open use (remembered for next time; each model type defaults to its own subfolder of models/) and open any model saved there.
Autosave and Restore	Every change is autosaved in the background (five rotating backups in the autosave folder). If the app is reopened with unsaved work from an earlier session, a Restore banner offers to bring it back. There is no unsaved-changes warning.

Decision Tree Editor

The Tree Editor draws the tree on a canvas. Branch labels, probabilities, costs and effects are edited on the canvas itself; node names are not shown. Strategies are the branches leaving the Decision node.

Feature	How it works
Strategy branches	A Decision-node branch only carries a label: it takes no probability, no $ and no E input. Cost and effect belong to the chance branches and terminals below it.
Path results on the canvas	Beside each terminal the canvas shows that path's probability p, total cost and total effect (branch increments plus the terminal value, with the terminal discounted by its year). Turn it on or off with 'Show path results'.
Strategy totals	Under each strategy node the canvas shows the base-case expected totals, labelled total_ plus the outcome symbol (for example total_$ and total_E). They equal the sum of p x value over that strategy's paths.
Base case results	The Paths table lists every path by branch labels (Strategy → label → label), with undiscounted and discounted cost/effect. Paths CSV and Trace CSV downloads give the step-by-step cumulative trace; a warning appears if the paths do not reconcile with the engine totals.
Cost-only models	Delete the Effect row in Outcomes (Model properties) to run a cost-only model: effect inputs are hidden and zeroed, and restored if an Effect row is added again. Analysis then skips the effect picker.
Text style	The toolbar above the canvas sets font, size and Bold / Italic / Underline separately for each text type (branch label, probability, cost, effect, path results, strategy totals). Settings are per text type, not synced.

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Feature	How it works
Save, Save as, Open	Save writes the current model straight to Folder/Filename, overwriting an existing file without asking. Save as downloads a copy through the browser. Open loads a model from the Saved models list.
Autosave and Restore	The model is autosaved in the background. If the app is reopened with an unsaved earlier session, a Restore banner offers to bring that work back.

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Model Calibration

Model Calibration adjusts a model's own Parameters and Variables so its output matches real-world data -- a known outcome rate, or, for Markov models, a Kaplan-Meier-style curve of state occupancy over time. It works the same way across model types: a sidebar button opens a panel below the model's Parameters/Variables/Distributions/Tables cards -- pick which inputs may move, enter the data to match, choose an algorithm, run it, then apply or discard. The Markov-specific target-row shape is covered in Part C; everything else on this page applies to both model types identically.

t row columns	Notes
State, Value, Weight	State is picked from the states of the strategy being ca occupancy fraction for that state at that cycle.
, Target ID, Value, Weight	Metric is total_cost, total_effect, or terminal_probability node and only applies to terminal_probability.

Algorithms

Algorithm	How it works / trade-off
Local Optimization	Steps from each input's base value toward the smallest mismatch until it converges. Fastest, but can settle in a local best fit.
Random Search	Draws random candidate combinations within Low/High bounds, keeps the best, optionally refines with Local Optimization.
Grid Search	Same idea as Random Search but over an evenly spaced grid. Cost grows fast with the number of calibrated inputs.
Bayesian (SIR)	Sampling-Importance-Resampling: draws from each input's prior, scores by fit, resamples into a posterior.
Bayesian (MCMC)	Metropolis-Hastings chain of proposed candidates, accepted or rejected by fit, approximating the posterior.

Bayesian Calibration

SIR and MCMC report a posterior mean, standard deviation, and 95% credible interval for every calibrated input, instead of one best-fit point. The result panel also reports a diagnostic -- effective sample size for SIR, acceptance rate (aim for roughly 15–50%) for MCMC -- alongside a converged/needs-review flag.

Applying a result

'Apply calibrated values' (Local/Random/Grid) or 'Apply posterior mean as base case' (Bayesian) writes the result into the model's real Parameters/Variables; 'Discard result' clears the run instead. A Bayesian result adds 'Apply posterior to PSA (recommended)', which fits a Beta distribution (when the posterior mean is strictly between 0 and 1) or a Normal distribution otherwise, and upserts it into that input's PSA Parameters row -- carrying the calibration's own uncertainty into every future probabilistic run instead of a single point estimate.

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

PSA Setup

Probabilistic Sensitivity Analysis draws each configured input from a distribution on every iteration instead of holding it at its base value. PSA Parameters is the table of inputs varied this way: each row targets either a Parameter/Variable by name, or one shape/scale slot of a Distribution, and carries its own PSA distribution type (Beta, Normal, Gamma, ...) and values (for example mean, se).

Correlation

Two PSA Parameters rows can be linked with a pairwise correlation coefficient so they no longer vary independently across iterations. This is for a simple two-way relationship; for three or more parameters that must move together under a shared structure, use Correlated Distribution instead.

Correlated Distribution

A Correlated Distribution group draws several PSA Parameters rows jointly from a single named distribution, instead of independently. Two kinds exist in DECIDE: Dirichlet and Multivariate Normal. A decision tree has no Markov transitions, so only Multivariate Normal applies here -- Dirichlet groups a Markov state's outgoing transitions and is covered in Part C.

Pick two or more existing PSA Parameters rows, set each one's mean and standard deviation, and fill in the pairwise correlation between every pair of members. An inconsistent correlation matrix is automatically repaired to the nearest valid one before sampling.

Applying a group generates one set of jointly-drawn values per PSA iteration and stores them against the group; the group panel shows Applied / Not applied so it is clear when a change still needs to be applied before the next PSA run picks it up. A group only affects the PSA iterations that come after it is applied -- the model's own base-case values are unchanged.

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Value of Information (EVPI/EVPPI)

The PSA results panel (shared across the decision tree and Markov model types) reports the Expected Value of Information alongside the CE plane, CEAC/CEAF, and convergence diagnostics — the standard HTA/payer measure of how much uncertainty is costing the decision, and whether further research is worth funding.

Metric	What it shows
EVPI	The per-patient value of resolving all PSA uncertainty at once, plotted across a grid of willingness-to-pay thresholds centred on the model's own WTP setting.
EVPPI by parameter	EVPI broken down by PSA parameter via polynomial regression of each strategy's net monetary benefit against that parameter's sampled values (the Strong/Oakley method). The regression degree is chosen per parameter by cross-validation and shown in the table's Degree column.
EVPPI for a group	Select 2+ parameters to estimate their joint EVPPI — the value of a study resolving all of them together, not generally the sum of their individual EVPPI, since parameters can interact.

Population EVPI/EVPPI

EVPI/EVPPI are per-patient by default. The Population inputs — annual eligible population, time horizon, discount rate — scale every value up to the total population the decision affects, using the same discounting convention the model's own cost calculations use: effective population = the sum, over each year of the horizon, of annual population / (1 + discount rate)^year, with the first year undiscounted. This is the number comparable to the cost of a proposed further study. Leaving the annual population at 0 shows only the per-patient values.

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Part C -- Markov Model

Engine variables

Variable	Meaning
_cycle	The model cycle, starting at 0. A decision tree evaluates once with _cycle = 0. In a Markov model, _cycle increases by one every simulated cycle.
_tenure	The number of consecutive cycles spent in the current Markov state, starting at 1. Decision trees evaluate with _tenure = 1.

Markov tenure states

Enable Tenure applied on a Markov state and set No. of tenure to model time spent in that state. DECIDE represents the state internally as temporary tunnel stages: entry starts at _tenure=1, each self-loop advances to the next tenure stage, and the final stage retains the self-loop so _tenure stays at its configured maximum. All transitions to another state are available from every tenure stage; a transition to another state restarts that destination state at _tenure=1.

Costs on a tenure-enabled state: Initial cost applies at _tenure=1 to everyone in the state in that cycle, including the portion that leaves during that same cycle (that portion receives only Initial cost, no Terminal cost); Recurring cost applies from _tenure=2 to the cohort portion remaining in the state; Terminal cost applies from _tenure=2 to the portion leaving the state next cycle (replacing recurring cost for that portion); at the time horizon, a cohort that remains receives recurring cost rather than terminal cost. To reproduce a plain per-cycle state cost c (as in TreeAge state rewards), enter c as Initial, Recurring and Terminal cost.

Markov Editor

Pending — the canvas mechanics for Markov (State and Transition editing, Strategies, absorbing states, tenure settings) have not yet been verified against the current app UI for this version of the manual and will be filled in a follow-up pass. See Engine variables above for tenure-enabled states, and Part B → File panel / Decision Tree Editor for the mechanics shared with the tree editor.

File panel

Same as Part B → File panel.

Model Calibration

Same mechanism as Part B → Model Calibration. The one difference is the shape of a target row: for a Markov model each row is a (Cycle, State, Value, Weight) point along the model's cohort trace, and Applying a result additionally shows a chart overlaying the calibrated model's trace against the target points — both already described in Part B.

PSA Setup

PSA Parameters and Correlation work exactly as in Part B → PSA Setup. Correlated Distribution's Dirichlet kind is available here, since a Markov model's states have outgoing transitions to group:

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Kind	Use for / draws
Dirichlet	A Markov state's outgoing transition probabilities, so they vary independently but always sum to exactly 1 on every PSA iteration. One row per outgoing transition, each a bare Parameter reference.
Multivariate Normal	Any two or more PSA Parameters rows that should move together under a general correlation structure (see Part B for detail).

Dirichlet: pick the Markov state; DECIDE finds its outgoing transitions automatically and requires every one of them to be a plain Parameter or Variable reference. A formula, or a transition using a complement / 'remaining probability' marker, cannot be part of a Dirichlet group, since the group's own draws already sum to 1 and the complement would then always resolve to 0. Set a concentration value (alpha) for each outgoing transition; alpha's relative sizes set the group's average split, and its overall scale sets how tightly the draws cluster around that split. If a transition is not yet a PSA Parameters row, DECIDE offers to add it with a starting Beta distribution before the group can be applied.

A Markov model's Dirichlet grouping applies to the currently open Strategy's transitions; it does not automatically extend to a state of the same name in a different Strategy.

Only Dirichlet and Multivariate Normal joint sampling are available for PSA today. A Multinomial-style joint distribution over discrete outcome counts is not yet implemented; it is a planned addition for a future model type where the outcomes it would apply to are counts of simulated events rather than continuous Markov transition probabilities.

Strategy comparison and ICER

Base case results list the strategies from the cheapest to the most expensive. The cheapest is the reference. Each of the others is compared with the **previous strategy that is not dominated**, not simply with the row above it.

Dominated: a strategy that costs more than a cheaper one but adds no effect. It is shown as *Dominated* and is never used as the comparator of the next strategy. **Extendedly dominated**: a strategy whose ICER against its comparator is higher than the ICER of the next, more effective strategy; it is shown as *Extendedly dominated*. Incremental cost and effect are measured against that comparator, and the ICER is their ratio. The same table is used for Decision Tree, Markov and Microsimulation models.

Value of Information (EVPI/EVPPI)

Same as Part B → Value of Information (EVPI/EVPPI). The PSA results panel and its EVPI/EVPPI reporting are shared unchanged between decision tree and Markov models.

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Part D -- Partitioned Survival Model (PSM)

A partitioned survival model (PSM) splits a cohort into three states -- **pre-progression** (progression-free), **post-progression** and **dead** -- directly from two survival curves, progression-free survival (PFS) and overall survival (OS). There are no transition probabilities: the proportion in each state at every time point is read off the curves, and costs and effects accumulate on those proportions. In DECIDE a PSM is a **PSM node** inside a decision-tree document, so the Decision node above it provides the strategies and everything Part B says about Parameters, Variables, Distributions, Tables, Sensitivity Analysis and PSA applies unchanged.

How a PSM node is calculated

For each cycle `c = 0, 1, ..., N` (N = Cycles), at elapsed time `t = c / cycles per year` (years):

State	Proportion of the cohort
Pre-progression	`S_PFS(t)`
Post-progression	`max(0, S_OS(t) - S_PFS(t))`
Dead	`1 - S_OS(t)`

Each state has a per-cycle cost and effect (the branch's cost/effect increment plus the value on its Terminal node). The cycle's reward is the sum over the three states of proportion x rate, discounted by `1 / (1 + rate)^t`, where the rate is the annual rate from Model properties and `t` is in years. Cycles `0 ... N-1` are summed; with **half-cycle correction** switched on, the first and last time points count half (cycle 0 weight 0.5, cycle N weight 0.5). Cost and effect rates are re-evaluated in every cycle, so they may use `_cycle` (for example a drug dose that stops after cycle 35: `if(_cycle<35, cDrug, 0)`). `_cycle` starts at 0; `_tenure` is always 1 in a PSM.

A curve's time axis is in the **time unit** you choose for it (year, month, week or day), so parameters can be typed exactly as published; DECIDE converts elapsed years to that unit before evaluating the curve. **Cycles per year** is separate: it sets how long a cycle is.

File panel

Same as Part B -> File panel. In addition, the PSM sidebar has an **Export code** panel (see "Export code" below).

PSM Editor

The canvas is the Tree canvas with one extra node type. A typical model is Decision -> one PSM node per strategy -> three state branches.

- **Strategies:** branches leaving the Decision node. A strategy can also lead to a Chance node whose branches end in PSM nodes.
- **PSM node:** select it on the canvas and use **Edit node** (right of the canvas) to set the PFS and OS curves, **Cycles**, **Cycles per year** and **Half-cycle correction**.
- **State branches:** a PSM node has exactly three branches with fixed roles -- Pre-progression, Post-progression and Dead -- each ending in a Terminal node. The branch's cost increment and effect increment, plus the Terminal's cost and effect, are the per-cycle rates of that state (Dead is usually 0). Use expressions with `_cycle` for time-varying costs; utilities for a cycle that is shorter than a year are entered per cycle (for example `0.804*3/52` for 3 weeks).

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

• **Parameters:** curve parameters and rates may be numbers or expressions that reference Parameters and Variables (Part A), so they can be varied in Sensitivity Analysis and PSA.

Survival distributions

Distribution	Parameters	S(t)
Exponential	rate	`exp(-rate*t)`
Weibull	shape, scale	`exp(-(t/scale)^shape)`
Log-normal	meanlog, sdlog	`1 - Phi((ln t - meanlog)/sdlog)`
Log-logistic	shape, scale	`1 / (1 + (t/scale)^shape)`
Gompertz	shape, rate	`exp(-(rate/shape)*(exp(shape*t) - 1))`
Generalized gamma	mu, sigma, Q (Prentice parametrisation)	regularised incomplete gamma; Q = 0 is the log-normal

`t` is in the curve's time unit. Parameters follow the flexsurv conventions described in Part A (Distribution types and parameters), so values fitted with flexsurv can be typed directly; values published in another convention (for example a Weibull given by a rate or a proportional-hazards form) must be converted with the table there first.

Fit from KM / patient data

You do not need the NMA module to get a curve. Open **Fit from KM / patient data** inside a curve's editor:

1. Choose the data type: **patient-level data** (rows of time and event: 1 = event, 0 = censored; paste or upload a CSV) or **digitised Kaplan-Meier points plus the numbers-at-risk table** (DECIDE reconstructs individual times with the Guyot algorithm).
2. Choose the time unit of the data and click **Fit all distributions**.
3. DECIDE fits all six distributions by maximum likelihood (right-censoring is handled) and lists log-likelihood, AIC, BIC and the largest gap between the fitted curve and the Kaplan-Meier curve, with a chart overlaying every fit on the Kaplan-Meier curve.
4. Pick a distribution and click **Use this fit**. The curve's distribution, parameters and time unit are filled in, and the fit (summary, number of patients and events, candidates, thinned KM points) is stored with the curve; it appears as the curve's **Source** in the Technical Report. Editing the curve by hand afterwards removes the stored fit.

Choosing between the candidates is a judgement: inspect the overlay and the long-term extrapolation, not AIC alone. The three panels below (Curve library, Extrapolation & hazard, Plausibility checks) are there for that. Piecewise or spliced extrapolation and hazard ratios/covariates are not available yet.

Curve tools: library, extrapolation and plausibility

Three panels work on the PFS and OS curves of a PSM node. In the desktop app they are in the **Setting** menu; in the browser they are sections of the sidebar. Each starts with a **Curve** selector (for example `psm_1 - OS`) that picks which curve of which PSM node you are working on. The same candidates feed all three panels.

Curve library. The library is a shortlist of candidate curves kept with one PFS or OS curve, so that you can compare several fits and choose the one used in the model, and keep a record of why.

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

1. The line *Curve in use* shows the distribution the model is currently using.

2. To fill the library, click **Add the N candidates from the last KM fit** (shown only after a *Fit from KM / patient data* run; it adds every fitted distribution with its AIC and BIC), or give the current curve a **Name** (for example "Weibull (best AIC)"), optionally write a **Rationale for choosing this curve**, and click **Save current curve as candidate**. A curve whose parameters are named model parameters rather than numbers cannot be saved as a candidate.

3. The table lists each candidate's name, distribution, parameters, time unit, AIC, note and whether it is **In use**. Select a candidate and click **Use this candidate** to copy its distribution, parameters and time unit into the curve (the model then runs with it), or **Remove from library**.

Saving a candidate under an existing name replaces it. The library is saved with the model file.

Extrapolation & hazard. Compares the current curve, the library candidates and the candidates of the last KM fit side by side over the model horizon. Enter the **Model horizon** and the **Observed follow-up** (taken from the stored Kaplan-Meier curve when there is one). The table gives, for every curve, survival at fixed landmark years, the median, the mean survival up to the horizon, the *share beyond data* (the part of that mean that lies after the observed follow-up, so it is extrapolation rather than data) and survival at the horizon. Two charts overlay survival (with the Kaplan-Meier curve in black and the end of follow-up dashed) and hazard per year.

Plausibility checks & background mortality. Lists warnings for each candidate: a fit that did not converge, a large share of the mean survival lying beyond the observed follow-up, many patients still alive at the horizon (so the mean is truncated), a Gompertz curve with a plateau, and (when a life table is entered) survival above the general population or a hazard below background mortality. For an OS curve it also warns if the PFS curve exceeds it. You can paste a **general-population life table** (one line per row: age, annual probability of death between 0 and 1) and the **Age at start**. DECIDE then compares each candidate with the general population and shows the mean survival with a background-mortality floor applied after a chosen year (from that year the hazard is the larger of the fitted hazard and general-population mortality). The floored curves are for comparison only; the model itself still uses the curve as typed.

Model properties

Outcomes (cost, effect and any extra rows) and the Discount table work as in Part B; the discount rates are annual and are applied to elapsed time in years, whatever the cycle length.

Validation

Before Run, DECIDE checks that every PSM node has its three state branches ending in Terminal nodes, that each curve has a known distribution and positive parameters, and whether PFS exceeds OS (a progression-free cohort cannot be larger than the cohort still alive). A missing branch, unknown distribution or non-positive parameter blocks the run until it is fixed. Curves that cross by a very small amount, which happens when PFS and OS are fitted independently, are reported as a review item (warning) in the self-checks and do not block the run.

Base case results

Base case results show, per strategy, total cost, total effect, incremental values and the ICER (CEA) exactly as in Part B. For paths ending in a PSM node the Paths table shows that node's discounted totals.

Survival Curves / State / Time Report

The **Report** button opens a panel for one PSM node:

- **Survival Curves:** PFS and OS by cycle.

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

- **State Report:** the node's totals split by state -- area under the occupancy curve (in cycles) and cost and effect for each state, undiscounted and discounted. Each cycle's own rates and the half-cycle weights are used, so the **Total** row equals the Run/ICER figure for the node.
- **Time Report:** one row per cycle with the three occupancies and the cycle's cost and effect.

Result Dashboard

The dashboard is the one described in Part B (Base case results, Tornado, PSA cards) with two extra PSM cards:

- **PSM - PFS / OS curves by strategy:** choose **All strategies** (one chart; colour identifies the strategy, solid lines are PFS, dashed lines are OS) or **Single strategy** (pick one strategy; blue is PFS, red is OS). **Curves** selects PFS and OS, PFS only or OS only. The choices are saved with the card.
- **PSM - State occupancy (stacked):** one panel per strategy showing the proportion pre-progression, post-progression and dead over time.

A new PSM model starts with Base case results and these two charts on its dashboard. When a strategy reaches several PSM nodes (for example under a Chance node) each node is drawn separately, labelled with the strategy and the node id. The dashboard exports to PDF.

Documentation exports

Under Base case results, **Documentation exports** produce material for review and for a technical appendix. They are built from the same calculation as Run, and DECIDE refuses to build them if its own replay of the model does not reproduce the engine's totals.

- **Model transparency Excel:** a workbook that recomputes the model with live formulas. Each PSM node has its own sheet ("PSM <node id>") with the curve parameters, survival, occupancy, state rates, discounting and the half-cycle correction by cycle, so changing a cell recalculates the node. The Tree Paths sheet links each path that ends in a PSM node to that sheet. Rate expressions that cannot be translated to an Excel formula are shown as amber cells with their resolved values and a comment. All six distributions are supported.
- **Technical Documentation Report (Word + Excel):** model structure, parameter table (parameters with no Source are highlighted), survival curves with their sources and curve-fitting evidence (AIC/BIC, fit to the Kaplan-Meier data), figures (survival curves with the KM data, overlay of strategies, stacked occupancy), occupancy and time-in-state tables, base case results and ICER, and the validation self-checks. The report is not built while a self-check of error severity fails.

Validation self-checks

Each PSM node is checked on every report, and the results are listed in the report:

Check	Meaning
PFS curve does not exceed OS curve	Review item (warning) if the curves cross
S(0) = 1, curves within [0, 1] and non-increasing	Each curve is a valid survival function
Occupancies sum to 1	Review item if curves cross

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Check	Meaning
Replay equals the engine	The cycle-by-cycle replay reproduces the node's cost and effect
Independent recomputation	A separate implementation of the survival formulas and the sum reproduces the totals
Zero reward gives zero	With all rates 0 the totals are 0
Dead-only reward	A reward only in Dead equals the sum of the (1 - OS) discounted
Reward 1 in every state	The total equals the sum of the discount factors

Failures of the error-severity checks stop the report; "Review" items are model-quality notes and do not.

Export code

The **Export code** panel in the PSM sidebar writes the model as a stand-alone script in **Python, R, Stata (Mata) or JavaScript**. Each PSM node becomes one function that loops over the cycles exactly as described above and returns its discounted cost and effect, and the script prints each strategy's total cost and effect (and the ICER for two strategies). Survival functions are included in the script. Python and JavaScript output has been checked against DECIDE's results; R and Stata output follows the same logic but should be run once on your machine to confirm.

Limits: a generalised gamma curve must use the Prentice (mu, sigma, Q) form; an expression that depends on `_cycle` and cannot be written in plain arithmetic / `if()` stops the export with a message instead of being fixed at one value.

Model Calibration, Sensitivity Analysis, PSA and Value of Information

These work as in Parts B and C. Curve parameters, state costs and utilities, and `Cycles` inputs expressed through Parameters can be varied in Sensitivity Analysis and given distributions in PSA; correlated sampling and EVPI/EVPPI are shared.

Worked example

`samples/psm_cai2021_sample.json` reproduces the structure of a published pembrolizumab + chemotherapy versus chemotherapy model (Cai et al., Front Oncol 2021): two strategies, each with a PSM node (log-logistic and generalised gamma PFS, log-logistic OS, curves in months), 174 cycles of 1/17.39 year, half-cycle correction and 5% annual discounting, with drug costs that stop after fixed numbers of cycles using `_cycle`. Open it, press Run, then open Report, Dashboard and the documentation exports to see every feature above. The base case is close to, but not an exact match for, the paper's Table 3.

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Part E -- Microsimulation

A microsimulation follows **individual patients**, one at a time, instead of a cohort average. Each simulated patient starts in a health state, and at every cycle a random draw decides the next state from that patient's current transition probabilities. Costs and effects accumulate per patient, and the results DECIDE reports are the **means across patients**. Because each patient is simulated on their own, a microsimulation can carry history that a cohort model cannot (time in a state, prior events, patient-specific values), at the price of simulation noise and a longer run time.

A Microsim model is built on the same canvas as a Markov model (Part C): a Decision node fans out into **strategies**, each strategy has its own root **M** node, and health states and transitions hang below it. Parameters, Variables, Distributions and Tables use the expression language of Part A. What is specific to Microsim is described here.

How a Microsim is calculated

For each of the **N patients**, and for each cycle `c = 0, 1, ..., Cycles - 1`:

1. The patient's current health state, the engine variables (`_cycle`, `_tenure`) and the current value of every **Tracker** (see Trackers below) are set up.
2. The outgoing transition probabilities of the current state are resolved (a # complement takes whatever probability is left).
3. The state's **cost** and **effect** for the cycle are computed and discounted: `value / (1 + rate)^c` (cycle 0 is not discounted). With **half-cycle correction** switched on, half of the difference between the last and the first cycle's reward is added, as in the Markov cohort engine.
4. The next state is drawn at random from the resolved probabilities.
5. Tracker update formulas run, and the reset rules of a state the patient has just entered are applied.

The patient's totals are averaged over all patients to give the strategy's expected cost and effect.

- **Initial state.** Each patient's starting state is drawn from the **Initial probability** of the health states (they must sum to 1; one state may use #).
- **Random numbers.** Every patient has an independent random-number stream derived from the **Random seed**. The same patient number uses the same stream in every strategy (common random numbers), so the difference between two strategies is not inflated by sampling noise. The same model, N and seed always return exactly the same results.
- **Precision.** Results carry a Monte Carlo standard error (MCSE): the standard deviation of the per-patient values divided by the square root of N. Quadrupling N halves it. A microsimulation converges to the equivalent cohort model as N grows.
- **Tenure states.** A health state with **Tenure applied** works as in Part C (Initial, Recurring and Terminal cost). Because a single patient cannot know in advance whether it is about to leave the state, the cost blends the three components using the probability that the patient leaves in this cycle, exactly as the cohort engine does.

Health states and branch nodes

Only the **direct children of the strategy's M node** are health states. A patient occupies one of them for a whole cycle, and each has a **per-cycle Cost and Effect**.

Nodes drawn below a health state (chance branches and terminal nodes) are **not** health states. They are decision points inside a single cycle:

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

• A patient who reaches a branch node is resolved through it to a health state **within the same cycle**; no time passes.

• A **terminal node ends the cycle**: the patient is in the health state that the terminal stands for, and in the next cycle moves on from that health state. A terminal is therefore not absorbing. Use **Relabel as terminal** on the canvas to point a terminal at the health state it represents (a terminal with the same name as a health state is matched automatically). Only the real **Dead** state is marked as absorbing.

• Branch and terminal nodes have **no per-cycle cost or effect**. Their Edit state box shows only **Incre_$** and **Incre_E** (for example QALY): a one-time cost or effect, charged once, in the cycle in which the patient passes along the transition into that node. These are the same numbers as the Incre values on the canvas edges.

For example, in a Sick-Sicker model Sick has three branches (recover, progress, die) plus a "stay sick" terminal. A patient who "stays sick" ends the cycle in Sick, and next cycle starts from Sick again.

File panel

Same as Part B -> File panel. The model's N patients, Random seed and Store N sample patients setting are saved with the file. Autosave and the Restore banner work as for the other editors.

Microsim Editor

• **Toolbar:** auto layout, **Run**, **SA**, **PSA**, **Dashboard** and Model Calibration. Each opens its panel below the canvas.

• **Strategies:** as in Part C. Each strategy has its own health states and transitions; Parameters, Variables, cycles, discounting and N are shared by all strategies.

• **Canvas view: State Diagram** (the editable canvas) or **Patient Journey** (see below).

• **Edit state:** select a node, edit it on the right and press **Apply**. A health state has Name, **Cost** (or Initial / Recurring / Terminal cost when **Tenure applied** is on, with **No. of tenure**), **Effect**, **Initial probability** and **Absorbing**. A branch or terminal node has Name, Incre_$ and Incre_E (see above).

Model properties

Setting	Meaning
Number of cycles	Number of cycles simulated. The reported trace has one more row than the number of cycles because it starts at cycle 0. To reproduce a source that reports time points 0 to 30, use 31 cycles.
Cycles per year	Length of a cycle. Used for the discount unit and by rate/probability conversions.
Half-cycle correction	Applies the half-cycle correction described above. Leave it off to reproduce a model that does not use one.
N patients	Number of simulated patients. Default 1,000. Larger N gives a smaller MCSE and a longer run.
Random seed	Seed of the random-number streams. Changing it gives a different, equally valid sample.
Store N sample patients	How many patients (the first N by index) have their full cycle-by-cycle path recorded for the Patient Trace view. Can be as large as N patients. 0 turns it off.

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Setting	Meaning
Outcomes, Discount	As in Part B. Discount rates can be entered per cycle or per year.
Analysis	Analysis type, effect objective, baseline strategy and willingness-to-pay threshold, as in Part C.

Trackers

A **Tracker** is a number carried by each patient through the simulation, for example "number of events so far" or "time on treatment". It can be used in any probability, cost or effect expression of that patient.

Column	Meaning
Name	The tracker's name, used in expressions. Must be unique.
Initial value	The value at cycle 0.
Update formula (per cycle)	Evaluated once per cycle after the patient's transition; the result becomes the new value. For example `n_events + 1`. The formula sees the values from the start of the cycle.
Reset rules	`state=value, ...`: when a patient enters the named state, the tracker is set to that value.
In output	Include the tracker in the Tracker summary of the results.

Tracker reset timing chooses when a reset takes effect: **On entering the new state**, or **One cycle after entering**. Use **Add tracker** to add a row, the bin icon on a row to delete that row, and **Update** to confirm the table.

Resampling distributions

In the Distribution table, **Resample** says how often a random value is drawn: **Per model** (once for the run), **Per PSA iteration**, **Per patient** (each patient gets their own draw, for example patient-specific age or risk) or **Per cycle**. Per-patient and per-cycle draws are what let a microsimulation represent patient heterogeneity.

Validation

Before Run, DECIDE checks the model and reports the first problem found, which blocks the run until it is fixed. The most common messages are:

Message	What to check
`initial_distribution_sum`	The Initial probabilities of the health states must add up to 1 (at most one may use #).
`transition_sum` / `missing_transitions`	The outgoing probabilities of a state add up to more than 1, or a health state has no transitions and is not absorbing.
`unresolved_alias`	A terminal node does not lead anywhere. Relabel it as a terminal of a health state, or give it the same name as the health state it represents.

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

Message	What to check
`non_real_state_loop`	Branch nodes lead back to each other without reaching a health state.
`no_real_states`	The strategy has no health state directly below its M node.
`invalid_n_patients`	N patients must be at least 1.
`duplicate_tracker`, `duplicate_state`	Names must be unique.

Run and Base case results

Press **Run** to simulate every strategy with the full N and open **Base case results**:

- **Strategy comparison:** total cost, total effect, incremental cost and effect, and ICER against the baseline, exactly as in Part C.
- **Trace:** a line chart of the proportion of patients in each health state at each cycle, and **one table** with the same rows: the proportions, and the raw and discounted cost and effect of each cycle. The table can be downloaded as CSV or Excel. With several strategies, **Strategy to trace** chooses which one is shown.
- **Tracker summary:** mean, 10th, 50th and 90th percentile of each tracker's final value, and the distribution of per-patient total cost and effect.

How long a run takes depends mainly on N, the number of cycles and the number of strategies. On the Sick-Sicker example one strategy takes about 0.6 seconds at N = 1,000, 6 seconds at N = 10,000 and 18 seconds at N = 30,000; N = 100,000 takes about a minute per strategy. Use a small N while building the model and the full N for the final run.

Patient Journey

Choose **Patient Journey** under Canvas view to look at the simulated patients instead of the model diagram. A **Strategy** selector above the view chooses which strategy to show.

- **Population Flow:** how the population moves between health states from cycle to cycle, as straight flow bands. The view is drawn from a simulation of at most 5,000 patients so that it stays fast; a note under the chart says so. **Run** always uses the full N.
- **Patient Trace:** one line per patient showing the state they are in at every cycle. The path of up to **Store N sample patients** patients is recorded. The browser draws up to 100 evenly spaced patients; **Prepare CSV** then **Download patient paths (CSV)** gives the path of every stored patient.

The legend lists only the health states.

Documentation exports

Press **Build Technical Documentation Report** under Base case results to produce a Word report and a companion Excel workbook. It is built from the results already on screen; DECIDE also runs the checks below on the model. The report contains: model structure and, when you have clicked the camera button on the canvas, the diagram; simulation settings (N, seed, cycles, discounting); the parameter table (parameters with no Source are highlighted); expected cost and effect with their MCSE and the ICER; a comparison with the equivalent cohort model; the validation self-checks; and a few cycles of each strategy's trace (the full trace is in the Excel file, one sheet per strategy). Unlike

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

the Markov and Tree workbooks, the Excel file holds values, not live formulas, because a microsimulation cannot be recalculated in a spreadsheet.

Validation self-checks

Check	Meaning
Patients are conserved	The proportions across states add up to 1 in every cycle.
No negative state occupancy	No state ever has a negative share.
Absorbing states never lose patients	Once patients enter an absorbing state they stay.
Mean of per-patient totals equals the total	The reported cost and effect are the mean of the per-patient values.
Per-cycle values add up to the totals	The discounted per-cycle cost and effect sum to the totals (when half-cycle correction is off).
Fixed seed reproduces results	Two runs with the same seed give identical results.
Agreement with the cohort model	The microsimulation is within 3 Monte Carlo standard errors of the same model run as a cohort (Review item if it is not).

A failing check of error severity stops the report; "Review" items are model-quality notes and do not.

Sensitivity Analysis, PSA, Dashboard and Model Calibration

These panels work as in Parts B and C. Every scenario, iteration or calibration step **runs the whole simulation again**, so they take far longer than one Run: use a small N (for example 1,000) and few iterations while setting them up, and keep Cycles and the number of strategies in mind. A Distribution with Resample set to **Per PSA iteration** is drawn once per PSA iteration and shared by all strategies in that iteration.

Worked example

`samples/microsim_darth_sicksicker_sample.json` is the model a new Microsim editor opens with. It is the Sick-Sicker model of Krijkamp et al., Medical Decision Making 2018;38(3):400-422 (https://doi.org/10.1177/0272989X18754513): four health states (Healthy, Sick, Sicker, Dead), two strategies (No treatment, Treatment), a Parameters table (costs, utilities and probabilities entered by name), **31 cycles**, **no half-cycle correction**, 3% discounting of costs and effects, and N = 100,000. The paper's Table 2 reports the deterministic cohort model and the microsimulation; DECIDE gives:

Result	Publication (cohort model)	DECIDE (cohort model)
No treatment	$75,976 and 15.83 QALY	$75,976 and 15.839 QALY
Treatment	$141,623 and 16.40 QALY	$141,623 and 16.400 QALY
ICER	$116,901 per QALY	$116,901 per QALY

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.

The microsimulation of the same model agrees with these within Monte Carlo error: at N = 10,000 the paper itself reports $75,790 and $141,211, and at N = 100,000 $75,996 and $141,644. Open the example, press **Run**, then open **Patient Journey**, the **Technical Documentation Report** and **PSA**.

Copyright 2026 Trang Nguyen - DECIDE. All rights reserved.